Paper 062 — Cross-Domain Echo with BH Γ_{cross} Collapse

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May 2026

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim that Q-COMPUTE platforms physically realize black-hole conditions; the echo is at substrate-mechanism level.
2. It does **not** claim novel cross-domain empirical predictions beyond what each arc supplies.
3. It does **not** introduce new substrate primitives.
4. "Cross-domain echo" = a structural correspondence between the Q-COMPUTE $\mathcal{M}_{\text{cap}}$ saturation phenomenon and the BH horizon $\Gamma_{\text{cross}} \to 0$ decoupling phenomenon; both are V5-cross-chain-bandwidth phenomena at substrate level.

Abstract

The Q-COMPUTE multiplicity-cap collapse ($\mathcal{M}_{\text{cap}}(s) \to \mathcal{M}_{\text{crit}}$, Papers_053–060) and the BH horizon $\Gamma_{\text{cross}} \to 0$ decoupling-surface formation (Paper_039) are FORCED to be structurally echoing phenomena at substrate level. Both arise from V5 cross-chain correlation kernel saturation: Γ_{cross} collapse marks the substrate-level loss of cross-chain participation flux across the horizon; $\mathcal{M}_{\text{cap}}$ saturation marks the substrate-level loss of cross-chain correlation budget at the platform's coherence boundary. Both are **V5-cutoff phenomena** under different platform configurations. The echo is **structural** (A→position): no Q-COMPUTE platform realizes a horizon, but the substrate-level mechanism is shared.

1 Primitive Inputs and Paper-Specific Postulates

1.1 Primitives invoked (per Paper_087)

P04 (Bandwidth), **P10 (Rule-type primitive)** [V5], **P11 (Commitment-irreversibility)**.

1.2 Upstream dependencies

- **I-039:** Horizon as decoupling surface ($\Gamma_{\text{cross}} \to 0$, Paper_039).
- **I-047:** Hawking spectrum (Paper_047).
- **I-050:** Page curve (Paper_050).
- **I-053–060:** Q-COMPUTE arc (Papers_053–060).
- **I-090:** V5 (Paper_090).

1.3 Paper-specific postulates

P-V5-Shared-Mechanism: $\Gamma_{\text{cross}} \to 0$ collapse (BH) and $\mathcal{M}_{\text{cap}}(s) \to \mathcal{M}_{\text{crit}}$ collapse (Q-COMPUTE) are governed by the same V5 substrate mechanism — namely, the cross-chain correlation budget reaching a saturation boundary.

§2.5 Audit Table

#	Step	Label	Notes
1	$\Gamma_{\text{cross}} \to 0$ at BH horizon	I	Paper_0
2	$\mathcal{M}_{\text{cap}} \to \mathcal{M}_{\text{crit}}$ in Q-COMPUTE	I	Papers_
3	Both phenomena involve V5 saturation	P-V5-Shared-Mechanism	Postulat
4	BH horizon = substrate cross-chain bandwidth collapse	I	Paper_0
5	Q-COMPUTE wall = substrate cross-chain bandwidth saturation	I	Paper_0
6	Structural echo (not operational identity)	A→position	Compos
7	No new cross-domain empirical predictions	I	Each ar

2 The Cross-Domain Echo

2.1 BH side: $\Gamma_{\text{cross}} \to 0$

Paper_039: the BH horizon is defined as the substrate surface where cross-chain participation flux Γ_{cross} between inside and outside collapses to zero. This is a V5-saturation phenomenon: V5 cross-chain correlations cannot transmit across the horizon boundary.

2.2 Q-COMPUTE side: $\mathcal{M}_{\text{cap}} \to \mathcal{M}_{\text{crit}}$

Paper_058 (Class C plateau): as Q-COMPUTE platforms approach $\mathcal{M}_{\text{crit}}$, the V5 cross-chain correlation budget saturates. The substrate cannot support additional rule-type configurations without forcing individuation.

2.3 Same V5 mechanism

By P-V5-Shared-Mechanism, both are **the same substrate phenomenon** under different platform configurations:

- **BH:** large-scale gravitational platform; horizon formation = boundary V5 collapse.
- **Q-COMPUTE:** laboratory-scale platform; wall formation = budget V5 saturation.

The substrate mechanism is V5-cutoff-driven in both cases.

2.4 What this is NOT

The echo is **structural**, not operational. A Q-COMPUTE wall does not produce Hawking radiation; a black hole does not run Shor's algorithm. The shared content is the **substrate-level mechanism** — V5 cross-chain saturation forcing individuation / decoupling.

This is an **A→position** framing claim, not a derivation.

3 Falsification Criteria

- **F1:** Substrate evidence that Γ_{cross} collapse and $\mathcal{M}_{\text{cap}}$ saturation are governed by **different** substrate mechanisms — refutes P-V5-Shared-Mechanism.
- **F2:** Empirical detection that V5 finite bandwidth (Paper_090) is not the load-bearing element in either phenomenon — refutes the shared-mechanism claim.
- **F3:** Discovery of a Q-COMPUTE platform that crosses $\mathcal{M}_{\text{crit}}$ without V5 saturation — refutes the substrate identification.

4 Position Statement

BH Γ_{cross} collapse and Q-COMPUTE $\mathcal{M}_{\mathrm{cap}}$ saturation are **FORCED to share** a substrate-level V5 mechanism under P-V5-Shared-Mechanism. The echo is **structural** (A$\rightarrow$position): same substrate phenomenon, different platforms. Operational identity NOT claimed.